PSet 1: The Price of Housing

```
library(tidyverse)

data_file <- "fmr2027.csv"

# fips and state have leading zeros
fmr <- read_csv(
  data_file,
  col_types = cols(fips = col_character(), state = col_character())
)

fmr <- fmr |>
  mutate(
    metro = factor(metro, labels = c("nonmetro", "metro")),
    # fips ending in 99999 is a whole county, otherwise a town
    record_type = if_else(str_ends(fips, "99999"), "county", "sub-county")
  )

# 2 bedroom rows only
fmr_2 <- filter(fmr, bedrooms == 2)

# one row per HUD area and bedroom size
areas <- fmr |>
  group_by(hud_area_code, hud_area_name, metro, bedrooms) |>
  summarise(
    fmr = first(fmr),
    pop = sum(pop2024),
    n_records = n(),
    .groups = "drop"
  )
areas_2 <- filter(areas, bedrooms == 2)

# areas that cross state lines are kept once per state
state_areas <- fmr |>
  distinct(stusps, hud_area_code, metro, bedrooms, fmr)
```

```
state_counts <- fmr_2 |>
  group_by(stusps) |>
  summarise(records = n(), `HUD areas` = n_distinct(hud_area_code)) |>
  pivot_longer(-stusps, names_to = "unit", values_to = "count") |>
  mutate(stusps = fct_reorder(stusps, count, .fun = max))

ggplot(state_counts, aes(count, stusps)) +
  geom_line(aes(group = stusps), color = "grey70") +
  geom_point(aes(color = unit)) +
  scale_x_log10() +
  labs(x = "count", y = "stusps", caption = "Figure 1") +
  theme_minimal(12)
```

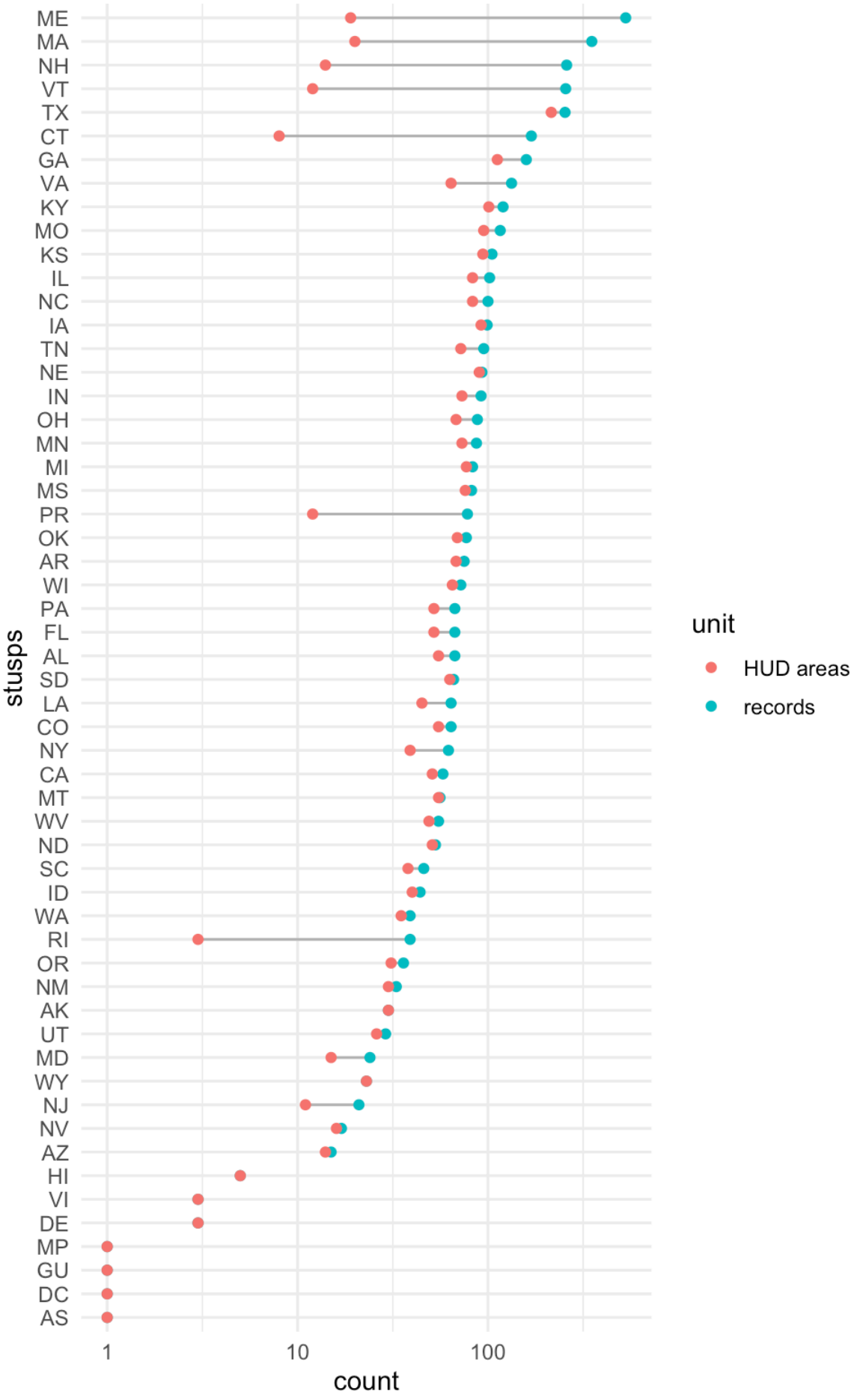

ME
MA
NH
VT
TX
CT
GA
VA
KY
MO
KS
IL
NC
IA
TN
NE
IN
OH
MN
MI
MS
PR
OK
AR
WI
PA
FL
AL
SD
LA
CO
NY
CA
MT
WV
ND
SC
ID
WA
RI
OR
NM
AK
UT
MD
WY
NJ
NV
AZ
HI
VI
DE
MP
GU
DC
AS
stusps
1
10
100
count
unit
HUD areas
records

Figure 1

```r
ggplot(areas, aes(fmr)) +
  geom_histogram(binwidth = 100, boundary = 0) +
  facet_grid(rows = vars(bedrooms), labeller = label_both) +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "count (HUD areas)", caption = "Figure 2") +
  theme_minimal(12)
```

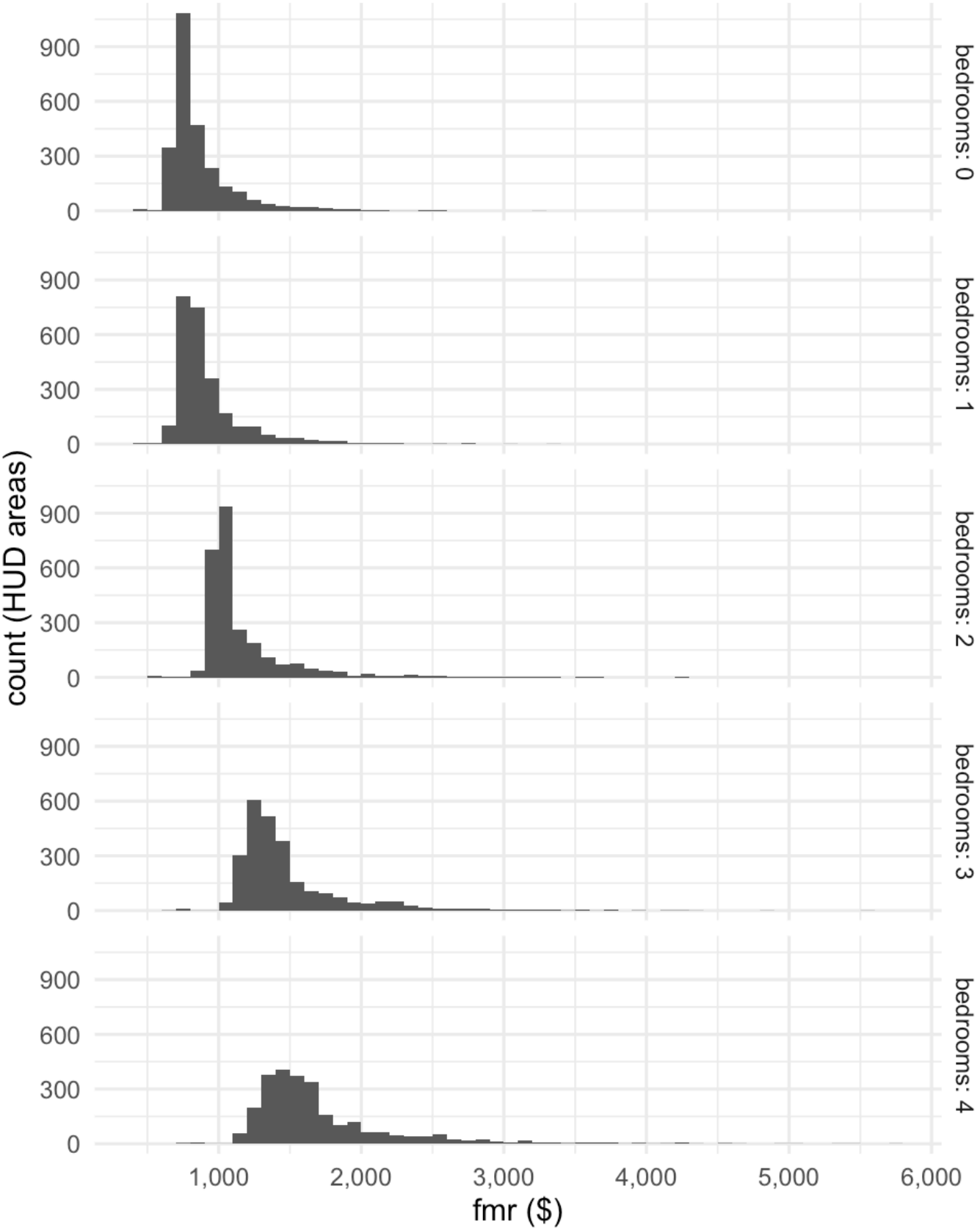


Figure 2

```
ggplot(areas, aes(factor(bedrooms), fmr, fill = metro)) +
  geom_boxplot(outlier.size = 0.8, outlier.alpha = 0.4) +
  scale_y_continuous(labels = scales::label_comma()) +
```

```
  labs(x = "bedrooms", y = "fmr ($)", caption = "Figure 3") +
  theme_minimal(12)
```

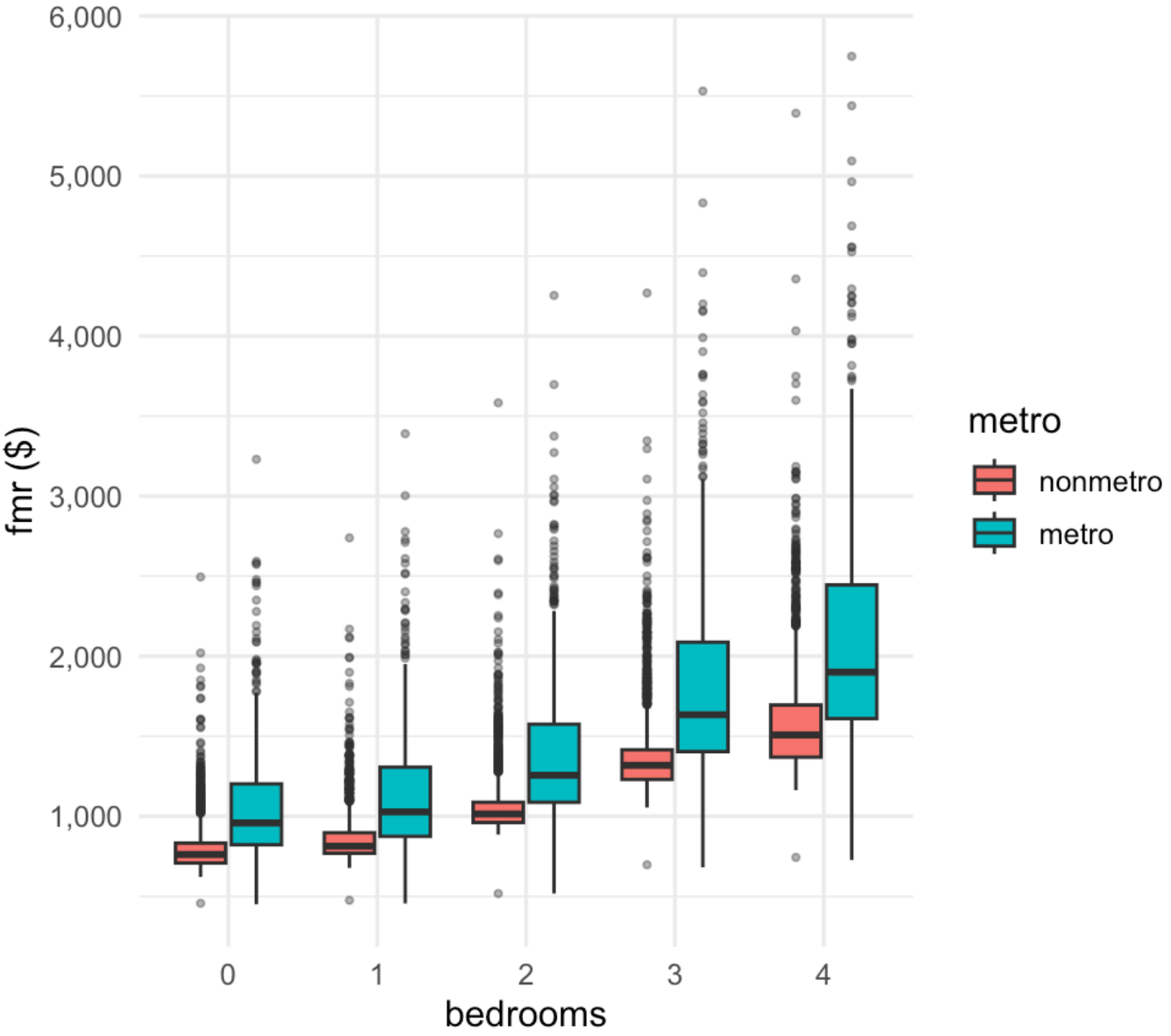


Figure 3

```
# ratio to the 2 bedroom fmr of the same area
ratios <- areas |>
  group_by(hud_area_code) |>
  mutate(ratio = fmr / fmr[bedrooms == 2]) |>
  ungroup() |>
  filter(bedrooms != 2)

ggplot(ratios, aes(ratio)) +
  geom_histogram(binwidth = 0.01, boundary = 0) +
  facet_wrap(~bedrooms, scales = "free_x", labeller = label_both) +
  labs(x = "fmr / fmr (2 bedrooms)", y = "count (HUD areas)",
       caption = "Figure 4") +
  theme_minimal(12)
```

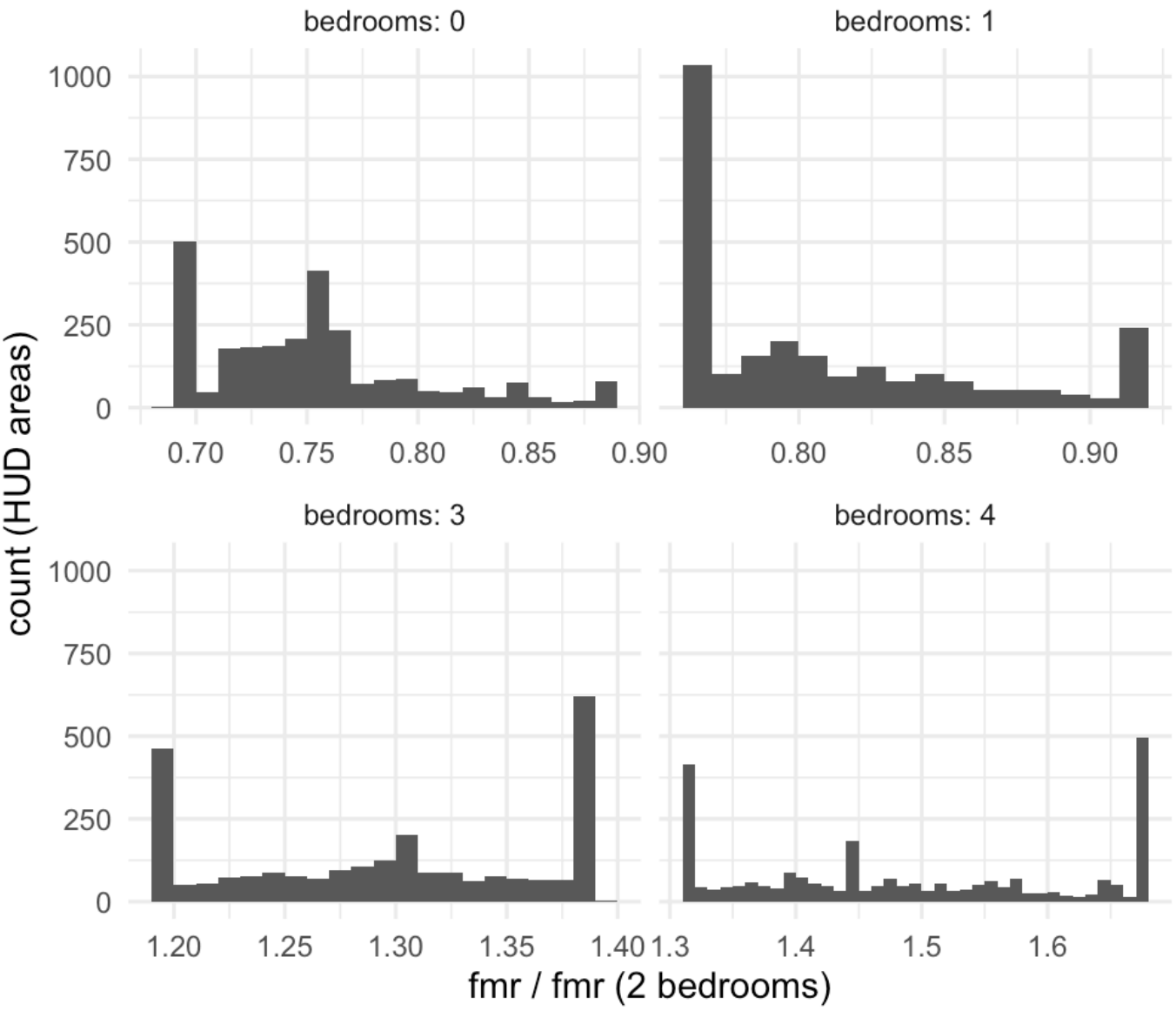


Figure 4

```
state_2 <- state_areas |>
  filter(bedrooms == 2) |>
  mutate(stusps = fct_reorder(stusps, fmr, .fun = median))

ggplot(state_2, aes(fmr, stusps)) +
  geom_point(
    aes(color = metro), alpha = 0.6, size = 1.2,
    position = position_jitter(width = 0, height = 0.25, seed = 1)
  ) +
  stat_summary(fun = median, geom = "point", shape = 124, size = 4) +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "stusps", caption = "Figure 5") +
  theme_minimal(12)
```

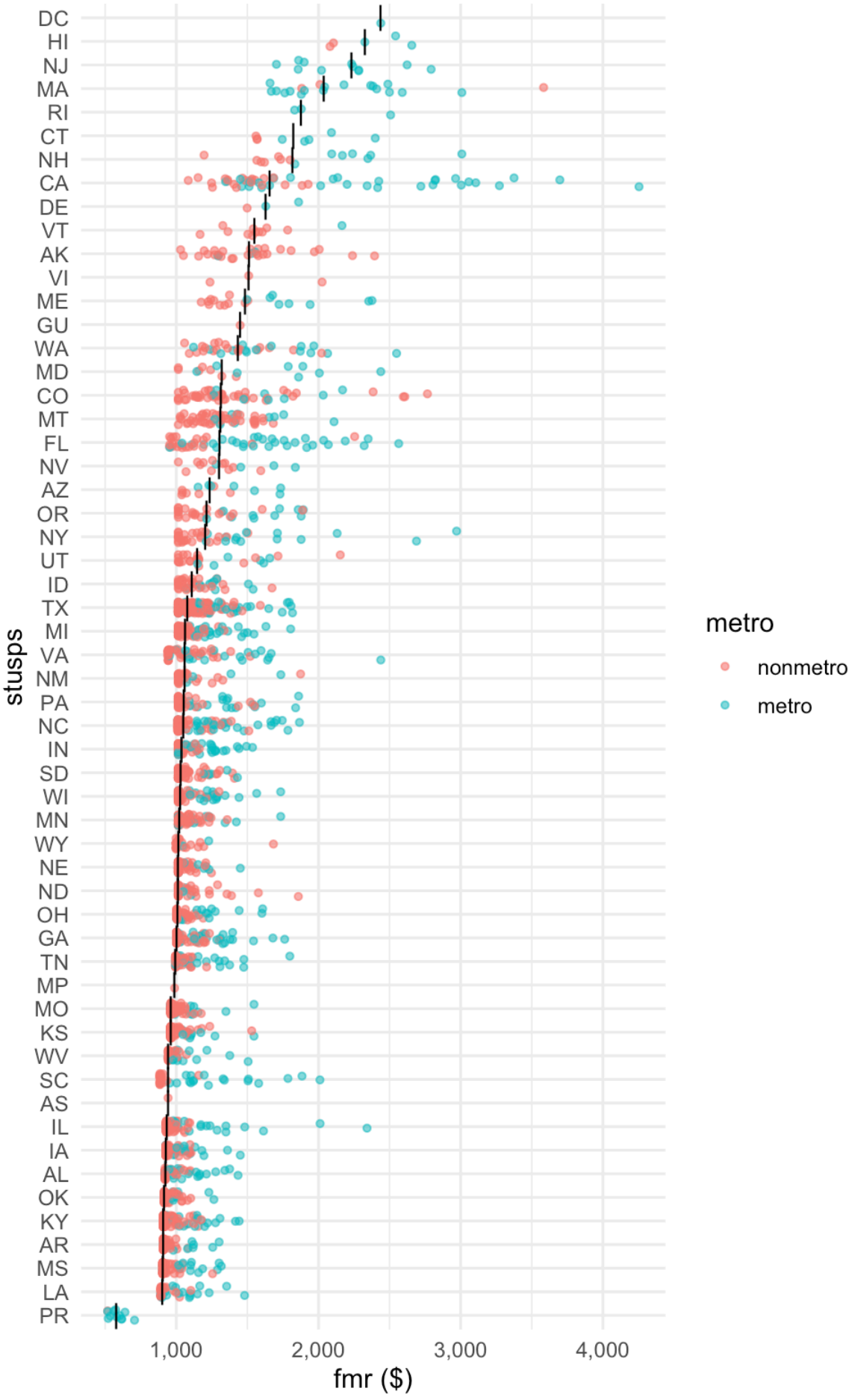

DC
HI
NJ
MA
RI
CT
NH
CA
DE
VT
AK
VI
ME
GU
WA
MD
CO
MT
FL
NV
AZ
OR
NY
UT
ID
TX
MI
VA
NM
PA
NC
IN
SD
WI
MN
WY
NE
ND
OH
GA
TN
MP
MO
KS
WV
SC
AS
IL
IA
AL
OK
KY
AR
MS
LA
PR
stusps
1,000
2,000
3,000
4,000
fmr ($)
metro
nonmetro
metro

Figure 5

```r
ggplot(areas_2, aes(pop, fmr, color = metro)) +
  geom_point(alpha = 0.4, size = 1) +
  scale_x_log10(labels = scales::label_comma()) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(x = "pop2024", y = "fmr ($)",
       caption = "Figure 6") +
  theme_minimal(12)
```

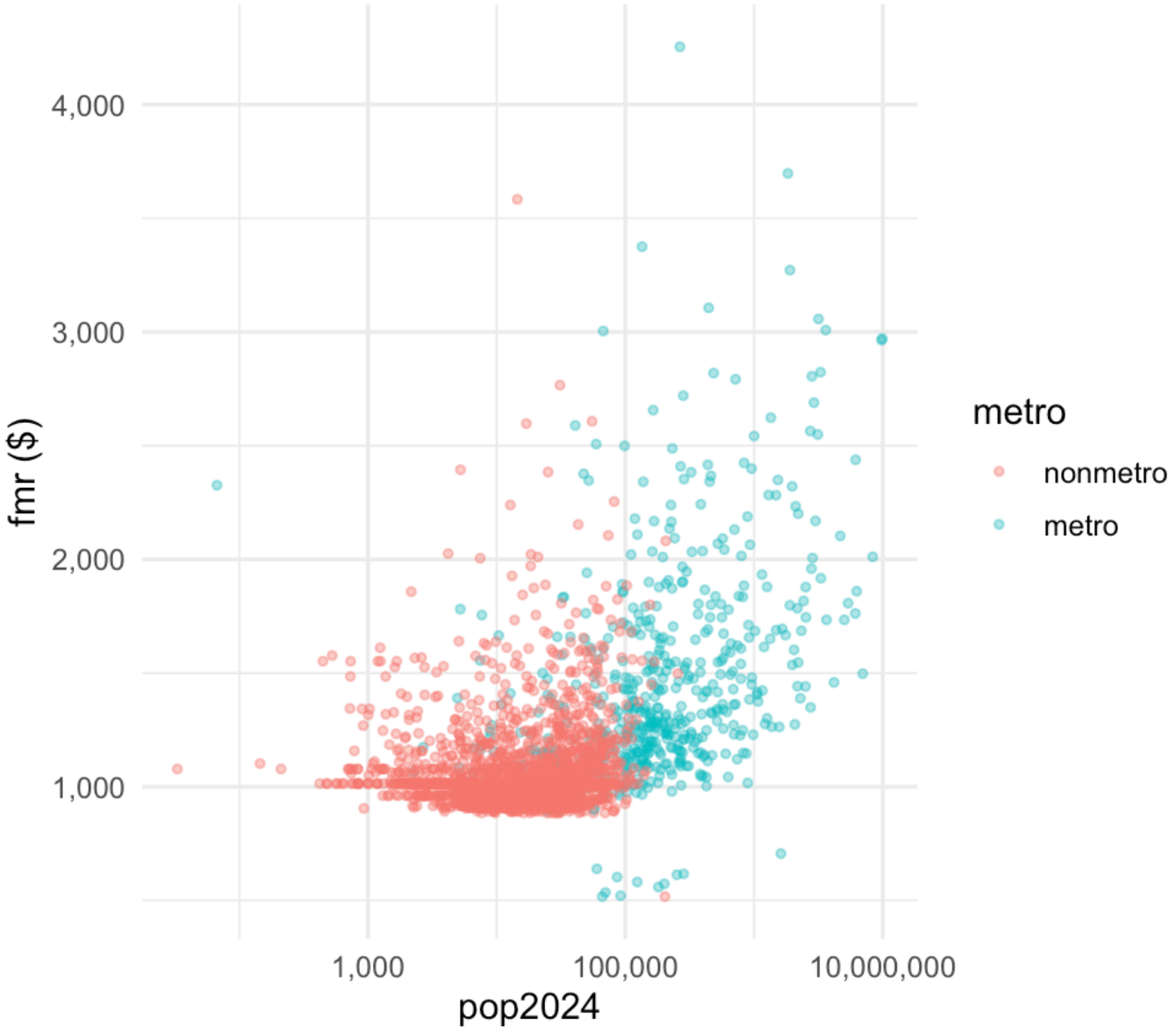


Figure 6

```r
# add 1 so the pop2024 = 0 records survive the log
ggplot(fmr_2, aes(pop2024 + 1)) +
  geom_histogram(bins = 40) +
  facet_grid(rows = vars(record_type), labeller = label_both) +
  scale_x_log10(labels = scales::label_comma()) +
  labs(x = "pop2024 + 1", y = "count (records)", caption = "Figure 7") +
  theme_minimal(12)
```

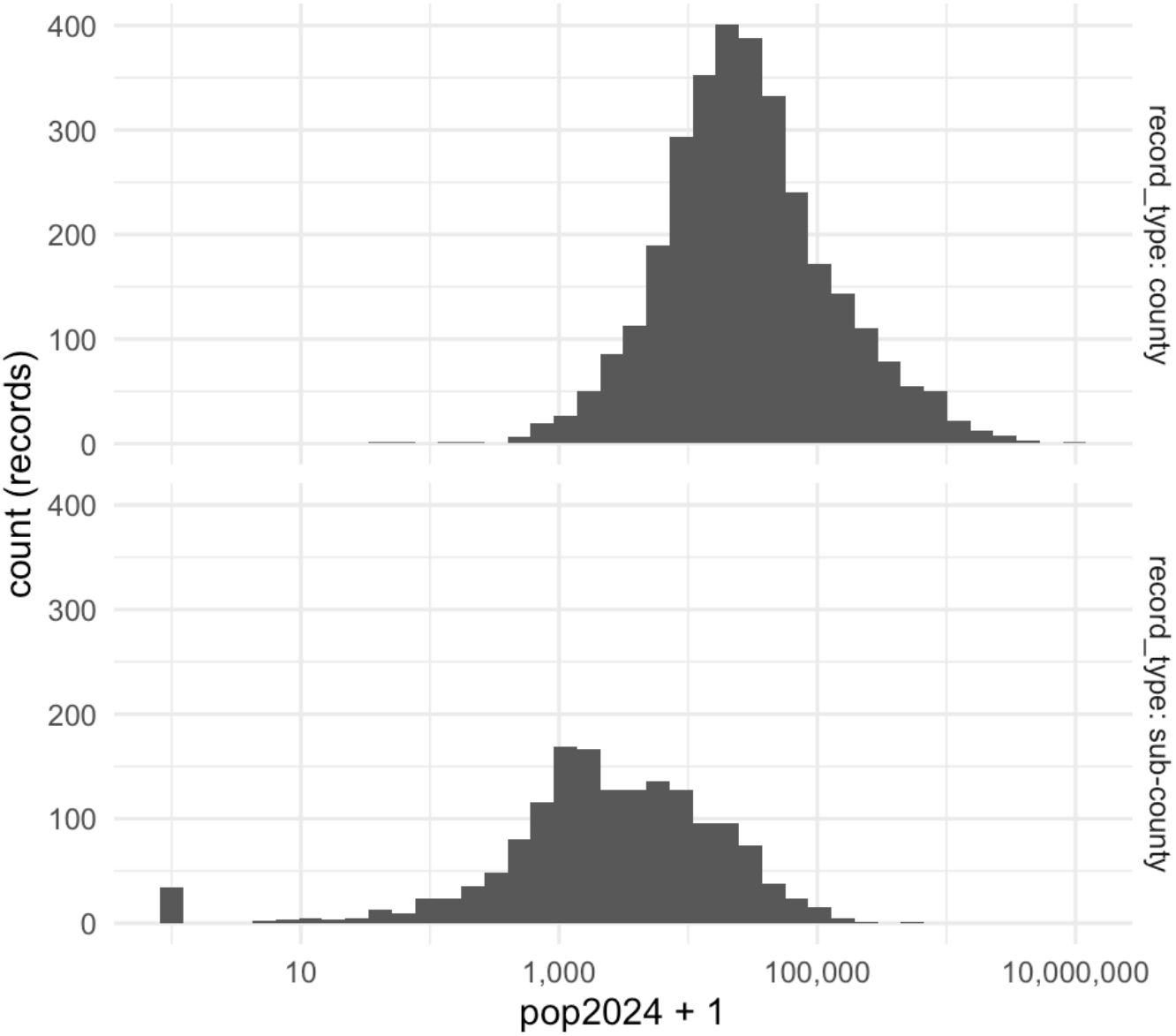


Figure 7

```
# 2 bedroom fmr by record and by HUD area
units <- bind_rows(
  transmute(fmr_2, unit = "records", fmr),
  transmute(areas_2, unit = "HUD areas", fmr)
)

ggplot(units, aes(fmr, color = unit)) +
  geom_density() +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "density", caption = "Figure 8") +
  theme_minimal(12)
```

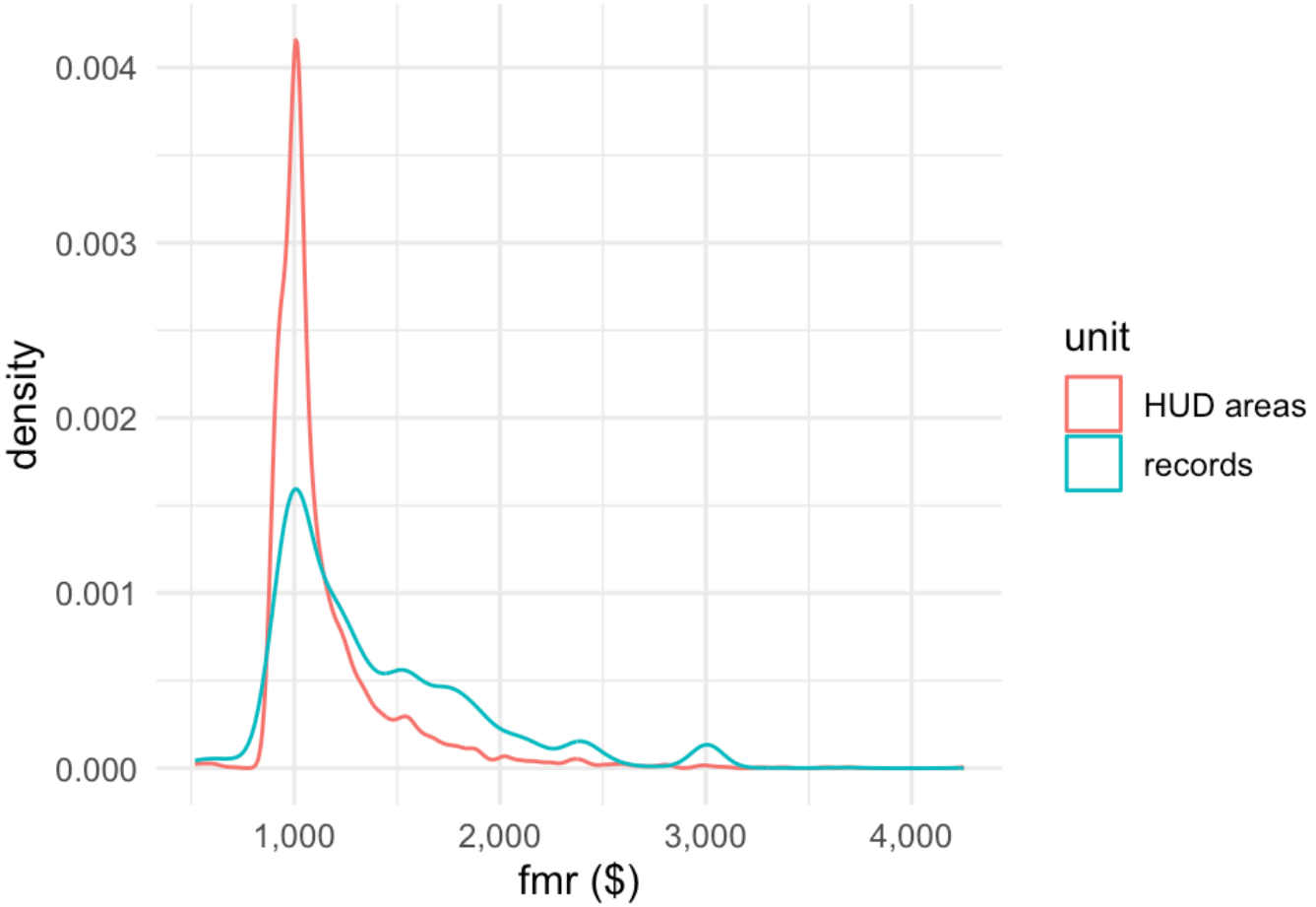


Figure 8

```
# 12 cheapest and 12 most expensive areas (2 bedrooms)
ranked <- arrange(areas_2, fmr)
extreme_codes <- c(head(ranked$hud_area_code, 12),
                   tail(ranked$hud_area_code, 12))

areas |>
  filter(hud_area_code %in% extreme_codes) |>
  mutate(
    hud_area_name = str_remove(hud_area_name, " HUD Metro FMR Area"),
    hud_area_name = fct_reorder(hud_area_name, fmr)
  ) |>
  ggplot(aes(fmr, hud_area_name, color = factor(bedrooms))) +
  geom_point(size = 2) +
  scale_color_viridis_d() +
  scale_x_continuous(breaks = scales::breaks_width(2000),
                     labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "hud_area_name", color = "bedrooms",
       caption = "Figure 9") +
  theme_minimal(12)
```

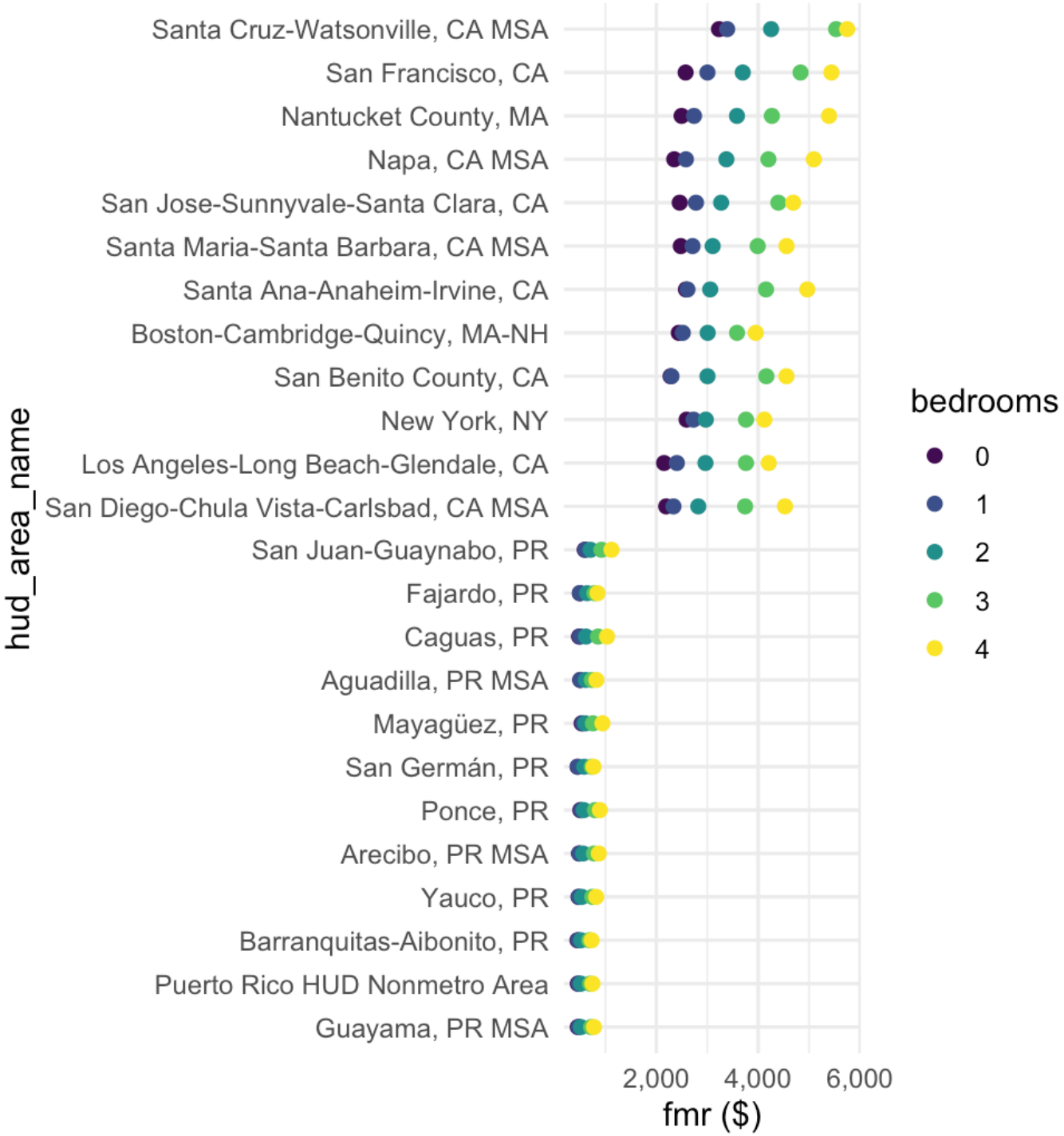


Figure 9

```
# areas with the most records
areas_2 |>
  slice_max(n_records, n = 15) |>
  mutate(hud_area_name = fct_reorder(hud_area_name, n_records)) |>
  ggplot(aes(n_records, hud_area_name)) +
  geom_point() +
  expand_limits(x = 0) +
  labs(x = "n_records", y = "hud_area_name", caption = "Figure 10") +
  theme_minimal(12)
```

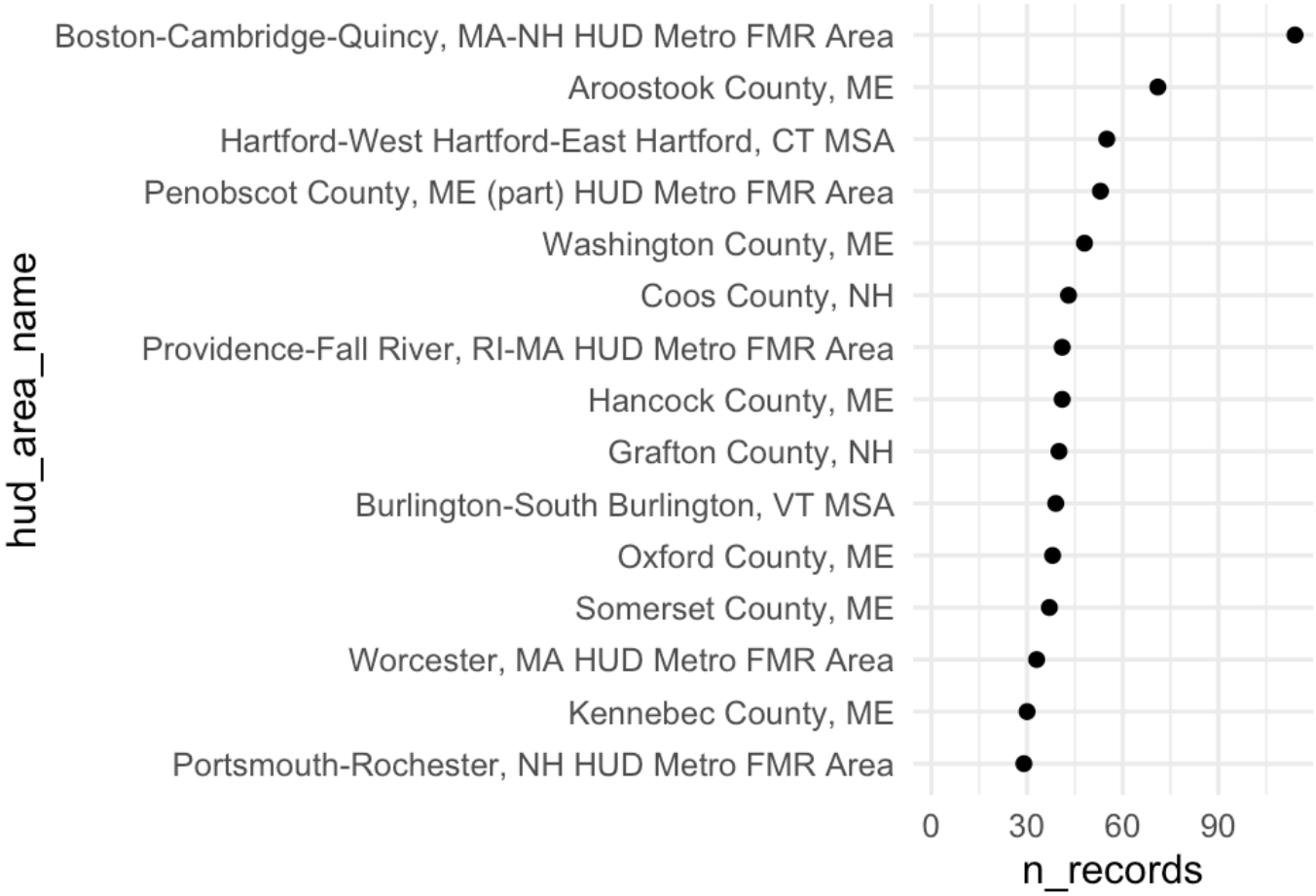

Boston-Cambridge-Quincy, MA-NH HUD Metro FMR Area
Aroostook County, ME
Hartford-West Hartford-East Hartford, CT MSA
Penobscot County, ME (part) HUD Metro FMR Area
Washington County, ME
Coos County, NH
Providence-Fall River, RI-MA HUD Metro FMR Area
Hancock County, ME
Grafton County, NH
Burlington-South Burlington, VT MSA
Oxford County, ME
Somerset County, ME
Worcester, MA HUD Metro FMR Area
Kennebec County, ME
Portsmouth-Rochester, NH HUD Metro FMR Area
hud_area_name
0
30
60
90
n_records

Figure 10